Lecture #18 - Properties of Determinants 3/21/26

Determinants $\longleftrightarrow$ Eigenvalues

↳ associated with every square matrix

$\det A = |A|$ → has as much info in 1 number

Invertible → when $\det A \neq 0$

singular → when $\det A = 0$

Properties

① $\det I = 1$ (Identity) → can a non-identity matrix have determinant of one? of course yes

② Exchanging rows → reverses sign of determinant

↳ det Permutation = 1 or -1

even odd

$\left\langle \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \right\rangle$

③a $\begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t \begin{vmatrix} a & b \\ c & d \end{vmatrix}$

③b $\begin{vmatrix} a+a' & b+b' \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}$

④ 2 equal rows → det = 0

↳ reasoning, use property 2 to prove: swap roles and signs swap but the # didn't change → has to be 0

⑤ Subtract $l \times$ row i from row k, the DET will not change

$\rightarrow \begin{vmatrix} a & b \\ c-la & d-lb \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a & b \\ -la & -lb \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} - l \begin{vmatrix} a & b \\ a & b \end{vmatrix}$] property 4

property 3b property 3a

$= \begin{vmatrix} a & b \\ c & d \end{vmatrix} - 0$

⑥ Row of zeros → $\det A = 0$

⑦ $U = \begin{bmatrix} d_1 & * & * & * \\ 0 & d_2 & * & * \\ 0 & 0 & \ddots & * \\ 0 & \cdots & 0 & d_n \end{bmatrix}$ What is det of upper Δ matrix?

$\det U = (d_1)(d_2)\ldots(d_n)$

= product of the pivots

↓

remove the * with row reduction → $\ldots d_2 d_1 \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \vdots & & \ddots & \end{bmatrix} \xrightarrow{\text{rule 1}} \det = (d_1)(d_2)\ldots(d_n)$

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⑧ $\det A = 0$ when A is singular $\rightsquigarrow$ row of zeros

$\det A \neq 0$ when A is invertible $\rightsquigarrow U \rightsquigarrow D \rightarrow d_1 d_2 \dots d_n$

⑨ $\det AB = (\det A)(\det B)$

then using 9. what's $\det A^{-1}$ $\rightarrow$ $A^{-1}A = I$

$(\det A^{-1})(\det A) = 1$

$\det A^{-1} = \frac{1}{\det A}$

ex. $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ $A^{-1} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/3 \end{bmatrix}$

ex. $\det A^2 = (\det A)^2$

$\det 2A = 2^n \det A$ $\leftarrow$ why? $\det 2A = 2^n \det A$ means taking 2 out of each row $\rightarrow$ 2^n for n rows

⑩ $\det A^T = \det A$

$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & c \\ b & d \end{vmatrix}$

quick proof: $|A^T| = |A|$

$\downarrow \quad \downarrow$

$|U^T L^T| = |LU|$

$|U^T||L^T| = |L||U|$

$\downarrow \quad \downarrow$

prop 7 prop 7

$\downarrow \quad \downarrow$

$1 \quad 1$

$U^T = U$